

CURRICULUM VITAE

Byol Kim

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Professional positions

Assistant Professor	Department of Statistics, Sookmyung Women's University	Sep 2024–(present)
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PAST POSITIONS

Postdoctoral Scholar	Department of Biostatistics, University of Washington	Sep 2021–Aug 2024
UW Data Science Postdoctoral Fellow	eScience Institute, University of Washington	Apr 2022–Aug 2024

Education

The University of Chicago, Chicago, IL	Aug 2021
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Doctor of Philosophy in Statistics

- Thesis title: *Uncertainty Quantification under Weak Assumptions*
- Advisors: Rina Foygel Barber and Mladen Kolar

The University of Chicago, Chicago, IL	Jun 2015
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Master of Science in Statistics

- Advisor: Lek-Heng Lim

Yale University, New Haven, CT	May 2009
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Bachelor of Arts in Applied Mathematics, *cum laude*

Inter-University Center for Japanese Language Studies, Yokohama, Japan	Jun 2008
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Completed the ten-month intensive program

Bibliography

PUBLISHED

1. Michael Hellstern, **Byol Kim**, Zaid Harchaoui, and Ali Shojaie. Spectral differential network analysis for high-dimensional time series. *The 28th International Conference on Artificial Intelligence and Statistics*, 2025. URL <https://openreview.net/forum?id=tuHVCBN5fw>
2. **Byol Kim** and Rina Foygel Barber. Black-box tests for algorithmic stability. *Information and Inference: A Journal of the IMA*, 12(4): 2690–2719, 2023. URL <https://doi.org/10.1093/imaiai/iaad039>
3. Ty O Easley, Zhen Ren, **Byol Kim**, Gregory Karczmar, Rina Foygel Barber, and Federico Pineda. Enhancement-constrained acceleration: A robust reconstruction framework in breast DCE-MRI. *PLOS ONE*, 16(10): 1–21, OCT 2021. URL <https://doi.org/10.1371/journal.pone.0258621>
4. **Byol Kim**, Song Liu, and Mladen Kolar. Two-sample inference for high-dimensional Markov networks. *Journal of the Royal Statistical Society: Series B*, 83(5): 939–962, 2021. URL <https://doi.org/10.1111/rssb.12446>

5. **Byol Kim**, Chen Xu, and Rina Foygel Barber. Predictive inference is free with the jackknife+-after-bootstrap. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems 33*, 2020. URL <https://proceedings.neurips.cc/paper/2020/file/2b346a0aa375a07f5a90a344a61416c4-Paper.pdf>

IN PREPARATION

6. Michael Hellstern, **Byol Kim**, and Ali Shojaie. Assumption-lean inference for spectral differential network analysis of high-dimensional time series.

Software

KLIEPInference.jl	https://github.com/mlakolar/KLIEPInference.jl
Enhanced-constrained acceleration	https://github.com/tyo8/ECA_Demo

Conferences and symposiums

INVITED SEMINARS

Statistical inference for high-dimensional spectral differential networks		
<i>Korean Statistical Society Summer Conference</i>	Gyeongju	Jul 2026
Black-box tests for algorithmic stability		
<i>International Day of Women in Statistics and Data Science</i>	online	Oct 2025
<i>Korean Statistical Society Summer Conference</i>	Gyeongju	Jun 2025
<i>Yonsei University BK21 Academic Seminar</i>	Seoul	Jun 2025
<i>Korean International Statistical Society Monthly Webinar</i>	online	Feb 2024
<i>Wayne State University Data Science Seminar</i>	online	Nov 2023
<i>Institute for Foundations of Data Science ML-Opt</i>	Seattle, WA	Dec 2022
<i>University of Washington Statistics Seminar</i>	Seattle, WA	Apr 2022
<i>University of Bristol Statistics Seminar</i>	online	Apr 2022
Jackknife+-after-bootstrap		
<i>International Seminar on Selective Inference</i>	online	Oct 2021

INVITED DISCUSSANT

<i>International Seminar on Selective Inference</i>	online	Nov 2022
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INVITED ORAL PRESENTATIONS

Two-sample inference for high-dimensional Markov networks		
<i>The 33rd New England Statistics Symposium</i>	Hartford, CT	May 2019

CONTRIBUTED POSTER PRESENTATIONS

Predictive inference is free with the jackknife+-after-bootstrap		
<i>Thirty-fourth Conference on Neural Information Processing Systems</i>	online	Dec 2020
<i>Institute for Foundations of Data Science Kickoff</i>	online	Sep 2020
Direct inference for Gaussian differential networks		
<i>The 4th Workshop on Higher-Order Asymptotics & Post-Selection Inference</i>	St. Louis, MO	Aug 2019
Two-sample inference for high-dimensional Markov networks		
<i>The 3rd Workshop on Higher-Order Asymptotics & Post-Selection Inference</i>	St. Louis, MO	Sep 2018
<i>The 2nd Midwest Machine Learning Symposium</i>	Chicago, IL	Jun 2018
<i>New Aspects of Statistics, Financial Econometrics & Data Science</i>	Chicago, IL	May 2018

MRI reconstruction

International Society for Magnetic Resonance in Medicine 27th Annual Meeting & Exhibition

Montréal, QC May 2019

American Association of Physicists in Medicine 60th Annual Meeting & Exhibition

Nashville, TN Jul 2018

Professional service

ORGANIZER

Topic-contributed paper session (1568, “Assumption-lean inference”)

2024 Joint Statistical Meetings Portland, OR Aug 2024

MANUSCRIPT REVIEW

Journal of Machine Learning Research 2023

Journal of the American Statistical Association 2023

NeurIPS 2022 Workshop on Distribution Shifts: Connecting Methods and Applications 2022

The 36th Conference on Neural Information Processing Systems 2022

Journal of the American Statistical Association 2022

The 10th International Conference on Learning Representations 2022

NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications 2021

The 37th Conference on Uncertainty in Artificial Intelligence 2021

The 24th International Conference on Artificial Intelligence and Statistics 2021

The 9th International Conference on Learning Representation 2021

The 34th Conference on Neural Information Processing Systems 2020

The 37th International Conference on Machine Learning 2020

The 23rd International Conference on Artificial Intelligence and Statistics 2020

The 33rd Conference on Neural Information Processing Systems 2019

Mentoring

Sookmyung Women’s University, Seoul, South Korea

Master’s Thesis Advisor Yoonjin Kim Mar 2025–Aug 2026

The University of Chicago, Chicago, IL

Co-advisor Student Inquiry & Research, Illinois Mathematics & Science Academy Oct 2014–Jun 2015

Teaching

The Korean Statistical Society, Seoul, South Korea

Invited Instructor Conformal Prediction by Examples Aug 2026

Sookmyung Women’s University, Seoul, South Korea

Instructor Conformal Prediction Mar–Jun 2026

Design and Analysis of Experiments Mar–Jun 2026

Mathematical Statistics II Sep–Dec 2025

Topics in Advanced Statistics I Mar–Jun 2025

Basic Statistics I Mar–Jun 2025

Mathematical Statistics II Sep–Dec 2024

The University of Chicago, Chicago, IL

Instructor	Statistical Models and Methods I	Jan–Mar 2019
Course Assistant	STAT 24410/30030 Statistical Theory and Method 1	Sep–Dec 2019
	STAT 20000 Elementary Statistics	Mar–Jun 2019
	STAT 24620/32950 Multivariate Statistical Analysis	Mar–Jun 2018
	STAT 22600 Analysis of Categorical Data	Jan–Mar 2018
	STAT 34700 Generalized Linear Models	Mar–Jun 2017
	STAT 34300 Applied Linear Statistical Methods	Sep–Dec 2016
	STAT 20000 Elementary Statistics	Mar–Jun 2016
	STAT 24400 Statistical Theory and Method 1	Jan–Mar 2016
Statistics Tutor	College Core Tutor Program	Oct 2014–Mar 2015

Honors, awards, scholarships

The University of Chicago Statistics departmental tuition scholarship	\$16,566	Jul 2014
The Richard U. Light Fellowship at Yale	\$32,640	Mar 2007
Bessemer Educational Scholarship	≈ \$200,000	Jul 2004

Skills

Programming	R, Python, Julia, MATLAB
Languages	English (fluent), Korean (native), Japanese (fluent)